AcceleratePSRO

The loop, the prior algorithm that already does half of it, and the words for what can go wrong

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> *"A world model is useful for more than prediction."*
>
> website, AcceleratePSRO, opening

This track has a vocabulary problem the other two do not. Almost every word in it belongs to somebody else already. PSRO, the restricted game, the response oracle, the strategy exploration problem, and regret all arrive with fixed definitions from empirical game-theoretic analysis. Dyna-PSRO already co-learns a world model with an empirical game, so the obvious sentence is taken. CyGym is somebody else's simulator with somebody else's game model attached.

What is mine here is small and it is worth stating exactly: three outputs, one experiment, and one number. This file separates the borrowed vocabulary from those, so that a sentence about this track cannot be read as a claim on the borrowed part.

> *"This is where "what if the world model is faulty" and "use the world model to expedite PSRO" turn out to be the same question asked twice."*
>
> website, AcceleratePSRO, the intuition

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1 Active

1.1 The loop

PSRO

The framework, from Lanctot et al. 2017, and the survey's statement of it.

> *"a framework for large games, known as Policy Space Response Oracles (PSRO), which holds promise to improve scalability by focusing attention on sufficient subsets of strategies"*
>
> Bighashdel et al. 2024, abstract

> *"In PSRO, a restricted game maintains a population of strategies (also known as policies, i.e., RL agents) and expands this population by introducing new strategies that respond to a specific distribution (mixture) of strategies within the existing population."*
>
> Bighashdel et al. 2024, Section 2.3

Definition: An iterative scheme that alternates two steps: solve the restricted game over the strategies collected so far, then compute a new strategy that responds to that solution and add it to the set.

Distinguishes it from: Self-play, which trains against the current copy and keeps a set of size one at every moment, and fictitious play, which responds to the empirical average of history rather than to a solution of a restricted game. PSRO generalizes both, and generalizes the double oracle method of McMahan et al. 2003 by replacing the exact best-response oracle with a learner.

Operational test: Ask what the new strategy responds to. If it responds to a mixture computed by solving a game over the strategies already collected, the loop is PSRO.

In my project: The loop is the object being accelerated. Nothing in this track proposes a new solution concept or a new oracle. The question is which of the loop's steps tolerate approximation and which do not.

the empirical game, the restricted game

Definition: The payoff model over a restricted set of strategies, with every entry estimated by simulation.

Distinguishes it from: The full game, whose strategy space is whatever the underlying decision process induces. The restricted game is a subset chosen by the loop, and the choice is the research problem rather than a preprocessing step.

Operational test: Count the entries. If the payoff of every joint strategy had to be estimated by running the game, the object is an empirical game and its size is the budget.

In my project: This is the term the two other tracks meet this one on. The Learn Structure page states the cost argument in one sentence: every payoff entry is estimated by simulation, already prohibitive beyond two players, so which strategies enter the empirical game governs what the analysis can deliver.

meta-strategy solver

Definition: The routine that computes a mixed strategy profile from the current restricted game, which is the target the next response is computed against.

Distinguishes it from: The solution concept it implements. The solver is a slot, and Nash is one filler among several, since a meta-strategy solver can implement other concepts.

Operational test: Ask what the response is aimed at. The output of this routine is that target.

In my project: One of the two per-iteration costs, and the second of the two the acceleration catalogue has to address. My own statement of the bottleneck names both: the per-iteration best-response solve and the meta-game solve.

response oracle

Definition: The routine that computes or learns a best response against the target the meta-strategy solver designated.

Distinguishes it from: An exact best-response oracle, which is what the double oracle method assumes and what almost no interesting game supplies. In PSRO the oracle is a reinforcement learner and its output is approximate.

Operational test: Ask whether the returned strategy is guaranteed optimal against the target. If it is only trained against the target, the oracle is approximate and every guarantee downstream of it inherits the approximation.

In my project: The first of the two per-iteration costs, and the one a world model is supposed to make cheap, since a response is trained on rollouts and rollouts are what a model can produce.

strategy exploration

> *"This challenge of restricted game construction with minimum computational cost (i.e., with the fewest strategies required) is described as the strategy exploration problem"*
>
> Bighashdel et al. 2024, Section 2.2

> *"the challenge of assembling effective subsets of strategies that still represent the original game well with minimum computational cost"*
>
> Bighashdel et al. 2024, abstract

And my own page states the same problem in my own words:

> *"assembling a strategy population that represents the full game under a computational budget"*
>
> website, AcceleratePSRO, the experiment

Definition: The problem of choosing which strategies enter the restricted game, so that the restricted game represents the full game well at the smallest number of members.

Distinguishes it from: Exploration in single-agent reinforcement learning, which chooses actions inside one episode. This chooses members of a population, and each member costs a full response computation.

Operational test: Ask what the budget is spent on. If it is spent deciding which strategies to add, the problem is strategy exploration.

In my project: Named on the page as the problem the track sits inside. The Learn Structure notes supply the reason it cannot be solved by care alone: the error from choosing a subset rather than the full policy space is not bounded, and no general guarantee says a chosen set is rich enough.

1.2 The prior algorithm

Dyna-PSRO

My page states the position without hedging.

> *"Co-learning a world model with the empirical game is not itself new"*
>
> website, AcceleratePSRO, the intuition

> *"Dyna-PSRO does exactly this."*
>
> same

The source, in its own words:

> *"We investigate the potential gain from co-learning these elements: a world model for dynamics and an empirical game for strategic interactions."*
>
> Smith and Wellman 2023, abstract

> *"Empirical games drive world models toward a broader consideration of possible game dynamics induced by a diversity of strategy profiles. Conversely, world models guide empirical games to efficiently discover new strategies through planning."*
>
> same

Definition: PSRO with two alterations: a world model trained in parallel with the loop's own routines on the data those routines already produce, and a Dyna-style learner used as the response oracle, so that planning in the model substitutes for part of the environment interaction a response would have cost.

Distinguishes it from: Model-based reinforcement learning applied inside one response computation and thrown away afterwards. Here the model persists across iterations and is fed by every episode the loop runs, which is what makes the co-learning claim rather than a per-iteration speedup claim.

Operational test: Ask where the model's training data comes from. If it comes from the payoff-estimation episodes and the response-learning episodes that the loop was going to run anyway, the model is being co-learned at no extra collection cost.

In my project: This is the baseline and the reason the track's question is narrower than it looks. Dyna-PSRO establishes that the loop is sound and that it reduces both regret and experience count on partially observable general-sum games. What it does not establish is what a faulty, adversarially stressed model does to the same loop.

> **What Dyna-PSRO settles and what my page asks**
>
> The source reports lower regret and substantially fewer experiences than PSRO, and its own account of model error is that prediction errors compound over long rollouts while medium-term planning still helps with an imperfect model.

My page asks a different question of the same loop: *Does model-based imagination still reduce the regret PSRO leaves on the table, or does a wrong model inject a bias that the game-solving step then amplifies?*
The difference is where the error comes from. Dyna-PSRO's model is imperfect because it is learned. The model this track studies is imperfect because something is stressing it, and the region it is wrong about is chosen rather than stumbled into. A result that model error is tolerable on average says nothing about a model error somebody selected.

1.3 The environment

CyGym

My page's gloss, which is the in-house sense:

> *"a simulation-based game-theoretic analysis framework for cybersecurity, which supplies the empirical game the loop is solved over"*
>
> website, AcceleratePSRO, environment

The source's own statement:

> *"We introduce a novel cybersecurity encounter simulator between a network defender and an attacker designed to facilitate game-theoretic modeling and analysis while maintaining many significant features of real cyber defense."*
>
> Lanier and Vorobeychik 2025, abstract

> *"Our simulator, built within the OpenAI Gym framework, incorporates realistic network topologies, vulnerabilities, exploits (including zero-days), and defensive mechanisms."*
>
> same

Definition: A two-party attacker and defender simulator with a formal simulation-based game-theoretic model attached, including a treatment of zero-day exploits and a PSRO-style method for approximating equilibria in the resulting game.

Distinguishes it from: A reinforcement learning environment that supplies only a transition function and a reward. CyGym supplies the game as well, which is why it is the environment this track needs and Daedalus is not. Daedalus is emulation and can carry a deployment claim, and it cannot carry a game solved to equilibrium.

Operational test: Ask whether the environment comes with a stated game model and a stated equilibrium method. If it does not, the loop has to invent both before it can be accelerated.

In my project: The two glosses agree and are scoped differently, and the difference matters. Mine says what CyGym supplies to me, the empirical game. The paper's says what CyGym is, an encounter simulator with a game model and a case study on the Volt Typhoon campaign. The paper's own PSRO-style method is therefore not a neighbour of this track. It is the thing this track is proposing to accelerate.

1.4 My own three outputs

"The working question is where a PSRO-like loop can be sped up without losing solution quality"

website, AcceleratePSRO, where acceleration can be introduced

acceleration lever

"a catalogue of the acceleration levers available in the loop"

website, AcceleratePSRO, the three outputs

Definition: A place in the loop where an expensive operation can be replaced by a cheaper one that the same solver can still consume.

Distinguishes it from: An implementation optimization, which makes the same computation run faster and changes nothing about what is computed. A lever changes what is computed and therefore has an approximation to account for.

Operational test: Ask whether the replacement changes the object the next step consumes. If it does not, it is engineering. If it does, it is a lever and it belongs in the catalogue with its shortcut named.

In my project: The catalogue is organized by the operation each lever attacks, and the existing PSRO variants supply most of the entries. Anytime PSRO attacks the response solve by keeping one approximate best response and updating it for a small number of steps per inner iteration rather than solving to convergence. Efficient PSRO attacks the meta-game estimation by a minimax formulation over an unrestricted-restricted game, which removes the need for meta-game simulation. The extensive-form double oracle line attacks the restricted game itself, by allowing population strategies to mix at every information set rather than only at the root. Dyna-PSRO attacks the interaction cost of the response. My own recipe question is the one that generated the catalogue: identify the operations that are the bottleneck of PSRO and reduce or bypass such operations.

risk register

"a risk register for the approximation shortcuts each lever takes"

website, AcceleratePSRO, the three outputs

Definition: The record, per lever, of what the shortcut assumes and what breaks when the assumption fails.

Distinguishes it from: A limitations section, which is written after the result and is about the study. This is written with the catalogue and is about the method, and it is what makes the catalogue usable by somebody choosing a lever.

Operational test: For each lever, name the quantity the shortcut is approximating and the direction the approximation moves it. A lever whose entry cannot name both is not understood well enough to be catalogued.

In my project: The entry that matters most is the model-based one, because its shortcut is the one with no bound. The transfer caution from Kiekintveld's line of work applies directly: a compact object is safe to substitute exactly when it preserves the property the solver needs, and the failure is borrowing a construction whose preserved property is not the one this solver needs.

validation protocol

> *“a validation protocol that says whether an accelerated run reached the same place a slower one would have”*

website, AcceleratePSRO, the three outputs

Definition: A procedure that decides whether an accelerated run’s answer is the answer the unaccelerated run would have produced.

Distinguishes it from: A speedup measurement, which compares wall clock or interaction count and says nothing about the answer. The protocol compares answers.

Operational test: Run both and compare the solutions on a quantity defined over the full game rather than over either restricted game, since two runs with different restricted games cannot be compared inside them.

In my project: The comparison quantity is regret against a combined strategy set, which is the standard device: pool the strategies both methods produced and evaluate each method’s solution against the pool. Without it, an accelerated run can look better simply by having explored a smaller set.

the crossover point

> *“The experiment settles it by degrading model quality along a controlled axis and measuring regret against interaction budget, relative to model-free PSRO.”*

website, AcceleratePSRO, the experiment

> *“The deliverable is the crossover point below which imagination stops paying.”*

same

Definition: The model quality at which model-based imagination stops reducing regret per unit of interaction relative to model-free PSRO.

Distinguishes it from: A threshold on model accuracy, which would be a property of the model alone. This is a property of the pair, model quality against interaction budget, and it moves with the budget.

Operational test: Degrade the model along the controlled axis and plot regret against interaction budget for both methods. The crossing is the answer, and if the curves do not cross inside the range studied, the answer is that the range was wrong.

In my project: This is the one number the track promises. It is a number about a specific loop on a specific environment, and it is worth being plain about that, because the temptation is to state it as a general fact about model-based game solving.

the negative case

> *“The risk is that the acceleration runs negative: model bias becomes bias in the best-response target, which the game-solving step then amplifies, and slower honest interaction wins.”*

website, AcceleratePSRO, the experiment

Definition: The outcome in which acceleration costs solution quality: model bias enters the response, the response enters the restricted game, and the solver compounds it.

Distinguishes it from: Slow convergence, which costs time and not quality. The negative case costs quality, and it is worse than the model-free run rather than merely no better.

Operational test: Compare the two methods' solutions on the combined strategy set rather than their speeds. A method that is faster and arrives at a worse profile is in the negative case.

In my project: This is the reason the track has a risk register at all, and the reason the deliverable is a crossing rather than a speedup factor. A speedup factor reported without the crossing is a claim about the wrong axis.

1.5 The measured quantities

regret

Definition: The payoff a player gives up by playing its solution rather than its best available response, taken over a stated set of alternatives, and summed or maximized over players for a profile.

Distinguishes it from: Exploitability, which is the same idea taken over the full strategy space rather than over a set. In this setting the full space is not available, which is why the set has to be named every time.

Operational test: Name the set the deviation is taken over. Two regret numbers computed over two different sets are not comparable, and comparing them is the most common way an accelerated run flatters itself.

In my project: The reported measure of the experiment, plotted against interaction budget. Learn Structure uses the word for a different quantity, regret per ordered pair of players, which is one of the two holes nothing in its measure set fills. Same word, different functional, and both are mine.

interaction budget

Definition: The number of environment steps consumed, counted across every routine of the loop.

Distinguishes it from: Compute budget. A model-based loop spends compute to save interaction, so a method that wins on one axis can lose on the other, and reporting only the axis it wins on is the thing the validation protocol exists to prevent.

Operational test: Ask whether the count includes the payoff-estimation episodes. It should, since in PSRO those are a large part of what the loop spends.

In my project: The horizontal axis of the experiment. Dyna-PSRO's own argument is stated on this axis too, that it requires substantially fewer experiences than PSRO, and the reason given is that collecting player-game interaction data is a cost-limiting factor.

imagination, imagined rollout

"imagined rollouts can stand in for expensive environment interaction when computing a best response inside PSRO"

website, AcceleratePSRO, the intuition

"When does imagination help, and when is it worse than honest, slower interaction?"

same

Definition: A trajectory generated by rolling the learned world model forward, used where an environment trajectory would have been.

Distinguishes it from: A simulation, which rolls a hand-written model forward. The word is reserved for the learned case because only the learned case can be wrong in a way nobody specified.

Operational test: Ask whether the generator of the trajectory was fitted. If it was, its error travels into the response and from the response into the restricted game.

In my project: The word honest in that second sentence is doing work and is worth keeping. Environment interaction is honest in the sense that it cannot be wrong about the environment, only expensive. That is the trade the crossover point prices.

2 Working

model-free PSRO

Definition: The comparison baseline: the same loop with the response oracle trained on environment interaction only.

Distinguishes it from: Rainbow and IQN, which are FOE-Dreamer's baselines. Those are policy learners and this is a solver, and the two tracks compare against different things because they claim different things.

double oracle

Definition: The predecessor of PSRO, in which each player's oracle returns an exact best response to the opponent's current restricted distribution, and the restricted game grows until neither oracle finds an improvement.

Distinguishes it from: PSRO, which replaces the exact oracle with a learner. Every guarantee that survives the replacement survives only up to the oracle's approximation error.

unrestricted-restricted game

Definition: The construction Efficient PSRO minimizes over, in which one side ranges over the restricted set and the other over the full policy space, which removes the need for meta-game simulation in the loop.

Distinguishes it from: The restricted game, where both sides are restricted. The asymmetry is the point, since it is what lets the target be computed without estimating a payoff matrix.

extensive-form double oracle

Definition: The variant in which the restricted game allows population strategies to mix at every information set rather than only at the root, which changes the iteration count from exponential in the number of information sets to linear.

Distinguishes it from: An improvement to the oracle or to the solver. This changes the restricted game's definition, which is a different lever and belongs in a different row of the catalogue.

strategy population

Definition: The set of strategies the restricted game is built over. Every member is kept, because dropping one changes the restricted game and therefore changes the equilibrium and the regret computed over it.

Distinguishes it from: The five other senses of population catalogued in `dict_population_based_training.te` in particular the hyperparameter sense the phrase was coined in, where the members are discarded and one model ships. The survey attaches that label to this loop and nothing carries across.

restriction error

Definition: The gap between the chosen strategy subset and the full policy space. It is not bounded, and no general guarantee says a chosen set is rich enough.

Distinguishes it from: Sampling error, the gap between estimated and true payoffs on a fixed set, which is bounded. Acceleration touches both, and only one of them can be reported with a bound.

3 Where the words disagree

> **Acceleration, on two different axes**
>
> The PSRO variants in the catalogue accelerate by reducing computation. Anytime PSRO cuts the inner-loop work per iteration. Efficient PSRO removes a simulation step. The extensive-form line cuts the iteration count.
> This track accelerates by reducing environment interaction, and pays for it in computation and in model error.
> The word covers both and the risks are not the same. A computational lever's worst case is that it converges more slowly. An interaction lever's worst case is the negative case above, a wrong answer arrived at quickly. Any catalogue that lists them in one column has to carry the axis in the row.

> **Faulty, in two senses that the page deliberately merges**
>
> On the Next page, a faulty model is one an adversary has found the wrong region of. On this page, a faulty model is one whose quality has been degraded along a controlled axis by me, in order to find a crossing.
> The two are not the same object, and the page says so by treating them as one question asked twice. The experimental version is the instrument and the adversarial version is the phenomenon, and the instrument is only a proxy for the phenomenon if the degradation axis reaches the region an adversary would have chosen. Whether it does is a design question for the degradation axis, and it is the question that decides whether the crossover point means anything outside the experiment.

Two pages carrying one paragraph

The intuition section of this page and the second section of the Next page carry the same text, down to the sentence order, differing in paragraph breaks and in one spelling of adversarially. A change to either has to be mirrored, or the duplication has to be resolved by making one of them the source and the other a pointer.

4 What crosses into the shared lexicon

Terms this track shares with the rest of Cyber World Modeling. The cluster sense is in `cwm_shared.tex` and the entries above carry only what AcceleratePSRO fixes about them: world model, imagination, interaction budget, regret, empirical game, and drift.

Two differ across projects and stay in both files. Regret here is the regret of a profile against a strategy set, while in Learn Structure it is a per-ordered-pair quantity that nothing yet measures. Population here is the strategy set the result is stated over, while in FOE-Dreamer it is the scripted user and attacker populations of the testbed, which are part of the environment and which no gradient touches.

5 Sources

- My own pages: `accelerate-psro.md` for the intuition, the three outputs, the experiment, the crossover point and the CyGym gloss; `next.md` for the duplicated paragraph and for the faulty-model question; `strategic-structure.md` for the cost of an empirical game.
- My own notes, in the Learn Structure folder: `Notes/Latex/sections/notes_0630.tex` for the recipe question about identifying the bottleneck operations, and for the strategy-space restriction question; the cards `Premise-KQ5-bottleneck.tex` for the two per-iteration costs, `Premise-KQ3-philosophy.tex` for the caution that a substituted compact object must preserve the property the solver needs, and `GenerateMDP-09-strategies-e-restrict.tex` for the restriction and sampling error pair.
- Lanctot, M., Zambaldi, V., Gruslys, A., Lazaridou, A., Tuyls, K., Perolat, J., Silver, D., and Graepel, T. A Unified Game-Theoretic Approach to Multiagent Reinforcement Learning. *Advances in Neural Information Processing Systems*, 2017. The paper PSRO is introduced in.
- Bighashdel, A., Wang, Y., McAleer, S., Savani, R., and Oliehoek, F. A. Policy Space Response Oracles: A Survey. *Proceedings of IJCAI 2024*, Survey Track, pages 7951 to 7961. Quoted from the abstract and from Sections 2.2 and 2.3.
- Smith, M. O., and Wellman, M. P. Co-Learning Empirical Games and World Models. arXiv 2305.14223, 2023; *Reinforcement Learning Journal*, RLC 2024. The paper Dyna-PSRO is introduced in. Quoted from the abstract; the two alterations and the compounding-error account are from the body.
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- Zhou, M., Chen, J., Wen, Y., Zhang, W., Yang, Y., Yu, Y., and Wang, J. Efficient Policy Space Response Oracles. arXiv 2202.00633, 2022, for the unrestricted-restricted construction.
- McAleer, S., Lanier, J., Fox, R., and Baldi, P. XDO: A Double Oracle Algorithm for Extensive-Form Games. *Advances in Neural Information Processing Systems*, 2021, for the extensive-form lever and its neural version.
- McMahan, H. B., Gordon, G. J., and Blum, A. Planning in the Presence of Cost Functions Controlled by an Adversary. *Proceedings of ICML 2003*, for double oracle.
- The four fields, the three tiers, and the rule that a term whose sense differs across projects stays in both files are from `_side/_lexicons/lexicon_method.tex`. The six senses of population are in `_side/_lexicons/dict_population_based_training.tex`.